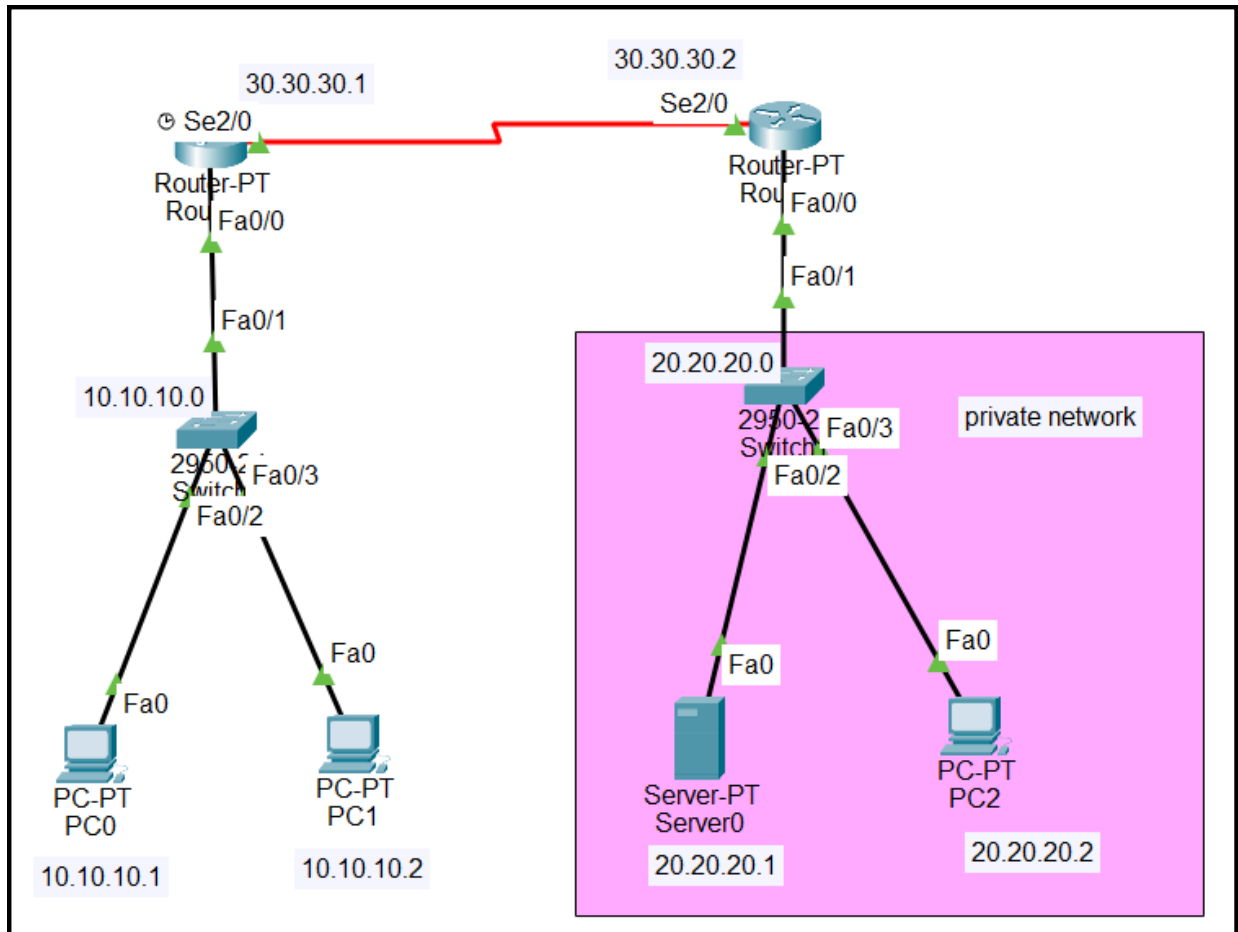


Practical-10

Examine Network Address Translation(NAT) in CISCO packet tracer



What is NAT?

NAT (Network Address Translation) allows private IP addresses to communicate with public networks by translating them into a public IP.

Step 1: Create Network Topology

Add devices:

- 2 Router
- 2 Switch
- 3 PCs
- 1 Server



Assign IP Addresses Inside Network (Private IP)

	IP Address	Default Gateway
Server0	20.20.20.1	20.20.20.0
PC2	20.20.20.2	20.20.20.0

Outside Network (Public IP)

	IP Address	Default Gateway
PC0	10.10.10.1	10.10.10.0
PC1	10.10.10.2	10.10.10.0

Step-2 Configure router

	Router0	Router1
Fa0/0	10.10.10.0	20.20.20.0
Se0/0	30.30.30.1	30.30.30.2

Step 3 Configure static routing



Router2

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Static Routes

Network

Mask

Next Hop

Add

Network Address

10.10.10.0/24 via 30.30.30.1

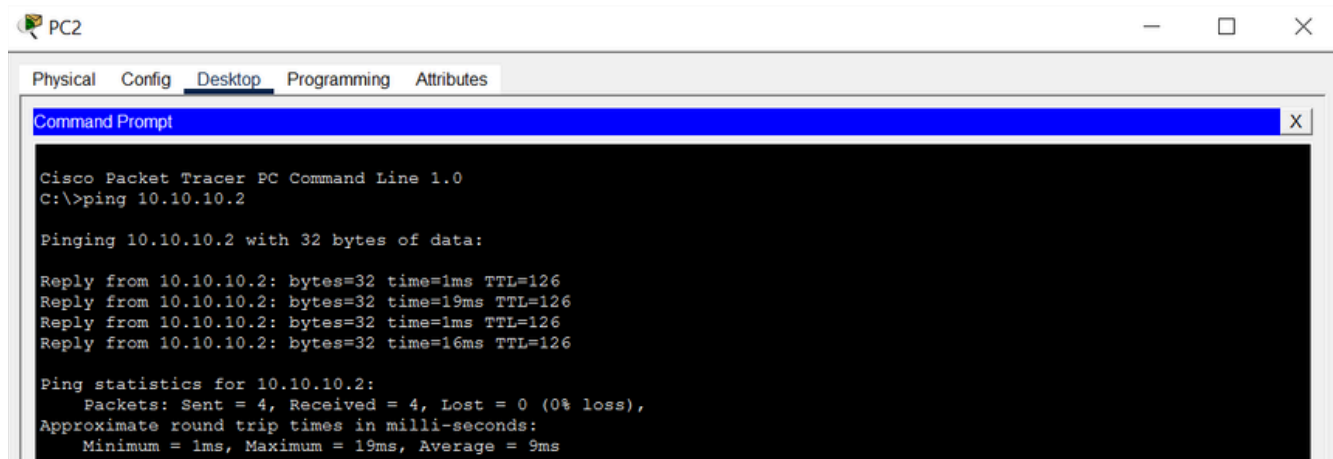
Step 4 Configure Router Interfaces

Go to CLI:

```
Router(config-if)#
Router(config-if)#ip nat inside source static 20.20.20.1 30.30.30.2
Router(config)#ip nat inside source static 20.20.20.2 30.30.30.2
Router(config)#interface fastethernet0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface serial2/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip route 10.10.10.0 255.255.255.0 30.30.30.1
Router(config)#
```

Step 5 Check connectivity

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC2	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC2	PC1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	Server0	PC1	ICMP		0.000	N	2	(edit)	(delete)



PC2

Physical Config **Desktop** Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=1ms TTL=126

Reply from 10.10.10.2: bytes=32 time=19ms TTL=126

Reply from 10.10.10.2: bytes=32 time=1ms TTL=126

Reply from 10.10.10.2: bytes=32 time=16ms TTL=126

Ping statistics for 10.10.10.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 19ms, Average = 9ms